

1.6 Detailed Specific Example: The Citarum River, Indonesia



The Citarum River lies on the Island of Java in Indonesia. The river flows 270 km from its source, located south of Bandung, to its mouth in the Java Sea, northeast of Jakarta. Water in the river has **domestic** (related to the home), agricultural, and industrial uses.



Figure 1.11: The Citarum River, Indonesia

Causes of pollution

In the 1980s, Indonesia began a period of rapid **industrialisation**. More than 2,000 factories, making products such as textiles and medicines, were built along the banks of the Citarum River. This was to take advantage of the supply of water. The factories routinely disposed of their waste in the river. **Domestic sewage** and garbage, such as plastics, entered the river from the towns along its banks. Agricultural runoff from fields also contributed to the decline in water quality.



Source A: Factories dispose of waste in the Citarum River



Source B: Domestic waste from homes adds to the pollution of the Citarum

TASK 1: Study Sources A B

- a Identify the source of water pollution that can be seen in each image.
- b Explain which source of pollution you think is the worst.

Impacts of pollution

The Citarum is known as one of the most polluted rivers in the world. In some places, the pollution is so bad that it can be difficult to see the water beneath the waste. Solid waste accumulates on the riverbed and clogs waterways, which can increase problems with flooding.

An estimated 9 million people live in close contact with the river, many of whom rely on the water for drinking, cooking, and washing. The water has high levels of bacteria from sewage, which cause illnesses such as **cholera**. It also contains heavy metals such as lead, aluminium, iron, and manganese, which can damage major bodily organs, such as the liver, kidneys, and heart. When used for irrigation, the water contaminates farmland.

Agricultural runoff, which is high in nutrients, leads to algae blooms and uncontrolled water hyacinth growth. These block sunlight and deplete oxygen levels in the water, making it difficult for aquatic life, such as fish, to survive. This threatens the livelihoods of communities that rely on fishing, and fish that are caught are contaminated with heavy metals and microplastics. Since 2008, the number of fish species in the Citarum has decreased by 60%.

REFLECTION

How does seeing photos like Source C help you to better understand the impacts of pollution on rivers? Which other topics related to rivers could you explore through real-life photographs to improve your understanding?



Source : Plastic waste on the Citarum River

TASK 2: Study Source

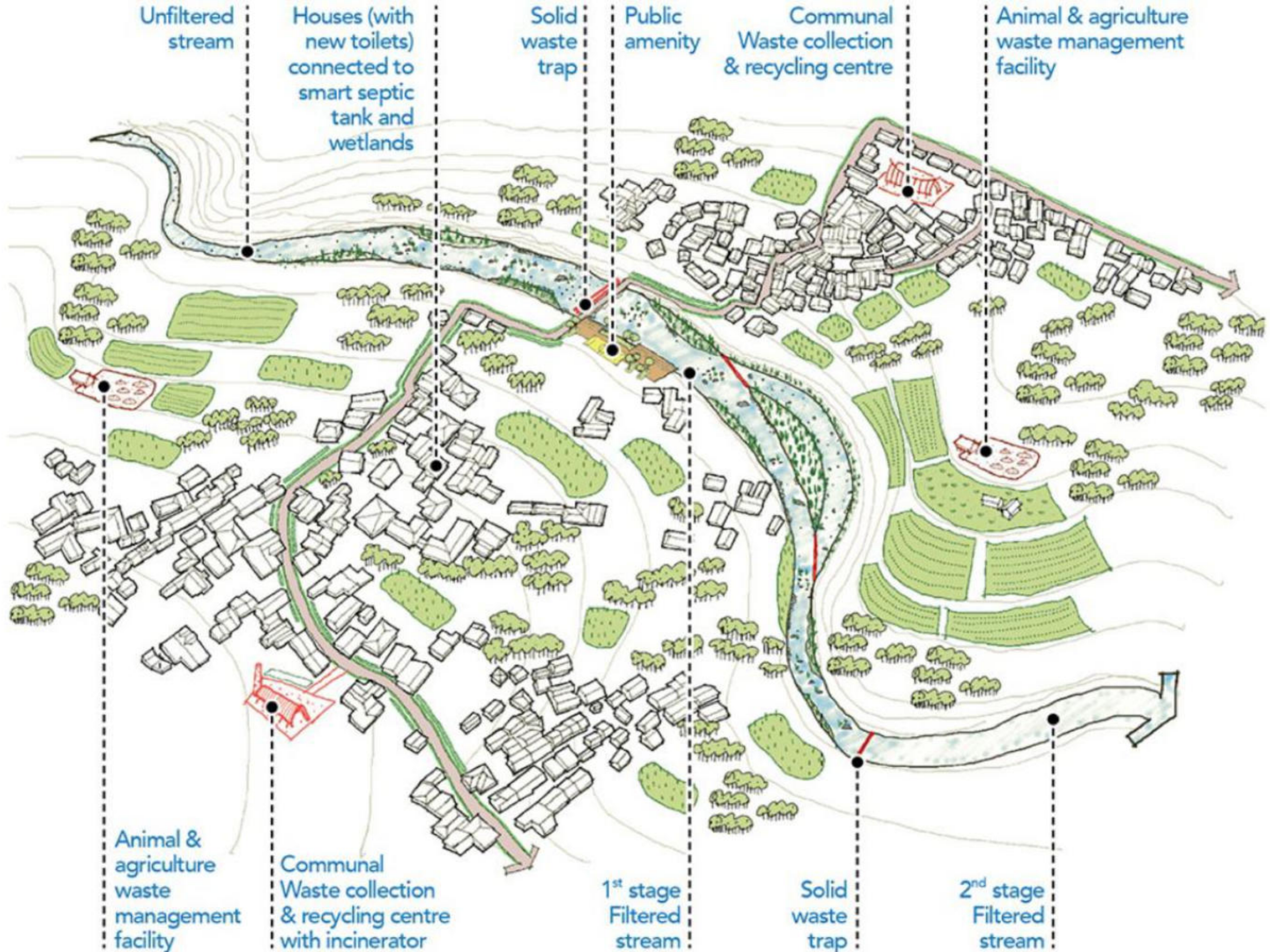
Suggest the impacts of plastic in rivers.

Sustainable management strategies and techniques

Efforts have been made to restore the Citarum River and its surrounding ecosystems. From 2018 to 2025, the Indonesian government established the Citarum Harum strategy with the goal of making the Citarum's water drinkable. In February 2023, the United Nations Development Programme (UNDP) carried out training for 40 companies operating along the river, to support the recognition and respect for access to clean water as a human right.

Many of the new techniques that have been introduced are sustainable, in that they balance environmental, social, and economic needs both now and in the future. They include:

- The introduction and enforcement of strict environmental laws. Authorities regularly monitor the river and issue fines to companies and individuals who continue to pollute the river.
- The creation of wastewater treatment and refuse facilities to prevent new contaminants from entering the river.
- The education of local communities about the dangers of pollution and the importance of preserving the river. Alongside Indonesia's military, locals have also been involved in clean-up efforts. Several communities now regularly collect and sell plastic waste for recycling. Community-led solutions empower local people to take control of their own health and well-being.
- A drive to encourage farmers to reduce their use of harmful chemicals. Buffer zones with native vegetation have been created, which filter pollutants from agricultural runoff.
- A more collaborative approach to reducing the problem of pollution. Government agencies and industries are beginning to work together to solve the problems of pollution.
- Research by Monash University and Universitas Indonesia to develop ways to detect and respond to pollution incidents. Designs include solutions to capture and treat wastewater and solid waste in villages along the Citarum.



Source **D**: Capturing and treating waste in villages along the Citarum

TASK 3: Study Source **D**

- a Identify which aspects of the design target:
 - i Agricultural pollution
 - ii Domestic pollution
- b The design in Source D does not feature measures to deal with industrial pollution. Suggest how industrial pollution could be targeted.
- c Compare these facilities with any that you are aware of in your country.

The sustainable management of the Citarum River is a complex and ongoing task, involving rules, infrastructure development, community involvement, and cooperation. While significant challenges remain, positive steps are being made to restore this vital waterway and to safeguard the wellbeing of the people and habitats that depend on it.

TASK 4

Using what you've learned about sustainable river management, work in groups to organise a one-day clean-up of a section of the Citarum River. You must include:

- a letter to locals, explaining why the task is necessary
- a risk assessment about the dangers that volunteers might face
- an equipment list, which includes tools to carry out the job, protective clothing and equipment
- a plan for how you will dispose of the collected waste safely
- a poster explaining why it is important to keep the area clean once your team have left.

IMPROVE THIS ANSWER

For a named river you have studied, explain the causes of pollution.

The pollution of the Citarum River is mainly caused by human activity. 1 Firstly, industrialisation along the riverbanks contributes significantly to pollution. 2 In the 1980s, more than 2,000 factories were built along the banks of the Citarum River. 3 Factories release untreated wastewater directly into the river, degrading water quality and harming aquatic life. 4

Extensive urbanisation in the surrounding area and farming practices also play a crucial role. 5 Furthermore, dumping of solid waste, including plastics and household trash, directly into the river adds to the pollution burden. 6

The pollution of the Citarum River in Indonesia could be resolved by introducing laws and building wastewater treatment plants. 7

1 The introductory sentence is strong because it sets the scene. Firstly, the question asks for a named example. The answer states an example, 'the Citarum River', although it could be clearer by locating the Citarum River. Secondly, it gives an overview of the causes of pollution, which the answer then goes on to discuss in more detail.

2 It is important to include key words. This answer does this well, as it uses words such as 'industrialisation' and 'urbanisation'. The answer also mentions farming – are there any other key terms that could be used instead of 'farming'?

3 The question has asked for place-specific references. The use of statistics, such as the number of factories along the Citarum, therefore makes this a stronger answer.

4 This part of the answer could be improved by adding specific information to give more depth to the explanation. For example, the untreated wastewater from the factories contains toxic chemicals, heavy metals, and industrial byproducts.

5 This bit of the answer reads like a list. Stronger answers will include developed statements, which means that more explanation is needed here. For example, urbanisation can cause pollution through sewage and domestic waste such as plastics. Pesticides, fertilisers, and sediment from agricultural runoff lead to algae blooms and uncontrolled water hyacinth growth, which block sunlight and deplete oxygen levels.

6 It is good to give examples as this increases the depth of the answer. Here, the answer gives examples of solid waste.

7 Although the final sentence includes accurate and good case study information, it does not answer the question. A better conclusion might be one that summarises the main causes of pollution.

Now, write an improved response to this answer using the guidance above.

SELF-EVALUATION CHECKLIST

After studying this topic, think about how confident you are with the different topics.
This will help you to find any gaps in your knowledge and learn more effectively.

I can:	Needs more work	Almost there	Ready to move on
understand the key characteristics of rivers and the underlying processes at work in drainage basins			
understand the processes involved in erosion, transportation, and deposition in the context of river systems			
describe and explain the formation of various river landforms found in the upper, middle, and lower sections of rivers			
explore the advantages and risks associated with living near river environments			
investigate and explain how regions with issues such as flooding and pollution have implemented sustainable strategies to address these challenges.			



> Topic 2

Coastal environments

LEARNING INTENTIONS

By the end of this topic you will be able to:

- explain how the processes of erosion, transportation, and deposition shape the coast
- describe the characteristics and explain the formation of coastal landforms
- identify the opportunities and hazards of living near coasts
- understand the causes and effects of coastal erosion and flooding
- assess hard and soft engineering strategies and techniques to overcome coastal flooding and erosion
- evaluate strategies and techniques used to protect the coast from the impact of tropical storms
- explain the importance of coral reefs and mangroves, and what threatens them
- evaluate the strategies and techniques used to overcome threats to coral reefs and mangroves.

GETTING STARTED

Working in pairs:

- 1 Identify as many physical and human factors and processes that affect the coast as you can.
- 2 Draw a concept map or diagram that shows how these factors are interconnected.

Share your ideas with the class.